

A disproof of Conjecture 00000000781

The closed-subgroup lattice of $1 + p\mathbb{Z}_p$

1 The conjecture and the reading

Definition: The lattice of closed subgroups of $1 + p\mathbb{Z}_p$ is the combinatorial structure of the multiplicative group. *Conjecture:* This subgroup lattice is isomorphic to a quotient lattice of the integers; the isomorphism is the valuation stratification; and the isomorphism preserves the stratified structure of the valuation filtration.

Reading. Neither the English nor the Chinese text restricts p (in particular, neither says “odd”), so the statement is read for every prime p ; we refute it at $p = 2$. The conclusion is a conjunction, so it suffices to refute the first clause, the existence of an isomorphism. The lattice is the set of closed subgroups of $U_1(p) = \{u \in \mathbb{Z}_p^\times : p \mid u - 1\}$ ordered by inclusion (the topology is the subspace topology from the unit group); a lattice isomorphism is the same as an order isomorphism.

Convention for “quotient lattice of the integers”. This is not a defined term. We formalize four readings (Lean: `C781.QuotientLatticeOfInt`): a lattice L that is a surjective lattice-homomorphic image of (a) the chain $(\mathbb{Z}, \leq)$, (b) the subgroup lattice of $\mathbb{Z}$, (c) its order dual (the divisibility lattice), or (d) a lattice isomorphic to the subgroup lattice of $\mathbb{Z}/n\mathbb{Z}$. Other senses are not covered, and the statement for odd p (where it is plausibly true) is not addressed.

2 Proof

Lemma 1 (Distributivity). *Every lattice in (a)–(d) is distributive.*

Proof. A surjective lattice homomorphic image of a distributive lattice is distributive. The chain is distributive. Every subgroup of $\mathbb{Z}$ is $n\mathbb{Z}$, with $n\mathbb{Z} \cap m\mathbb{Z} = \text{lcm}(n, m)\mathbb{Z}$ and $n\mathbb{Z} + m\mathbb{Z} = \text{gcd}(n, m)\mathbb{Z}$; the identity $\text{gcd}(\text{lcm}(a, b), \text{lcm}(a, c)) = \text{lcm}(a, \text{gcd}(b, c))$ is checked on prime factorizations, giving distributivity of the subgroup lattice of $\mathbb{Z}$, hence of its dual. For $\mathbb{Z}/n\mathbb{Z}$, pulling back along the surjection $\mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z}$ is an order embedding preserving meets and joins into the subgroup lattice of $\mathbb{Z}$. (Lean: `distrib_of_surj`, `dual_distrib`, `gcd_lcm_key`, `subZ_distrib`, `subZMod_distrib`, `quotient_distrib`.) $\square$

Lemma 2 (A diamond at $p = 2$). *Let $\rho : \mathbb{Z}_2 \rightarrow \mathbb{Z}/8$ be reduction and, for $a \in \{3, 5, 7\}$, $H_a = \{u \in U_1(2) : \rho(u) \in \{1, a\}\}$. Then H_a is a closed subgroup, $H_7 \cap H_3 = H_7 \cap H_5 = \{\rho = 1\}$, and the only closed subgroup containing H_7 and H_a ($a = 3, 5$) is $U_1(2)$.*

Proof. ρ is continuous (its fibres are balls of radius 2^{-3}) into a discrete space, so $H_a = \rho^{-1}\{1, a\}$ is closed; it is a subgroup because $a^2 = 1$ in $\mathbb{Z}/8$. Every $u \in U_1(2)$ has $\rho(u) = 1 + 2s$, so $\rho(u) \in \{1, 3, 5, 7\}$. If a closed subgroup X contains H_7 and H_a , then it contains -1 ($\rho(-1) = 7$); if $\rho(u)$ is neither 1, 7 nor a , then $\rho(-u) = a$, so $-u \in X$ and $u = (-1)(-u) \in X$. (Lean: `pi8_cont`, `Hres`, `top_of`, `isLUB_pair`, `isGLB_pair`.) $\square$

Lemma 3 (Main theorem). *There is no order isomorphism from the closed subgroups of $1 + 2\mathbb{Z}_2$ onto a lattice as in (a)–(d).*

Proof. Suppose e is one. An order isomorphism carries meets and joins (as greatest lower bounds and least upper bounds) to meets and joins, so by Lemma 2, $e(H_7) \wedge e(H_3) = e(H_7) \wedge e(H_5)$ and $e(H_7) \vee e(H_3) = e(H_7) \vee e(H_5)$. Since L is distributive (Lemma 1), cancellation gives $e(H_3) = e(H_5)$, so $H_3 = H_5$. But 3 is a unit of $\mathbb{Z}_2$ with $3 - 1 = 2$, so $3 \in H_3 \setminus H_5$. Contradiction. $\square$

3 Lean and audit

File `Conjecture781/Basic.lean` (identical to `Tlmc/C781.lean`). The statement block defines `U1`, `QuotientLatticeOfInt`, `Conjecture p` (there is a lattice L with `QuotientLatticeOfInt L` and an order isomorphism `ClosedSubgroup (U1 p)  $\simeq_o$  L`) and `Claim := not (forall p prime, Conjecture p)`. The theorem is `C781.main : C781.Claim`. Axioms: only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`.

Python check. The same diamond exists in $(\mathbb{Z}/8)^\times$: subgroups $\{1, 3\}, \{1, 5\}, \{1, 7\}$ meet pairwise in $\{1\}$ and join to everything (attempt file `check_code`).